

SAMBP Research Programme: Comprehensive Summary, Thesis Modification Plan, and Future Research Objectives

Post-OpenDSS/ANDES/pandapower Phase
Covering Technical Reports TR-56 through TR-66

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Document type	Internal Research Summary
Total TRs covered	TR-01 through TR-66
Phase 7 TRs (new)	TR-56, TR-57, TR-58, TR-59, TR-62, TR-65, TR-66
Journal papers	8 (Papers A–H)
Validation scenarios (cumulative)	~91
Overall selectivity (TR-66)	12/12 = 100%

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1 Purpose and Scope

This document provides a structured record of the SAMPB (Self-Adaptive Model-Based Protection) research programme from the point at which three external simulation tools were installed and activated: **OpenDSS** (via `opendssdirect.py`), **ANDES** (dynamic stability simulator for IBR), and **pandapower** (power network analysis).

Before these tools, all SAMPB validation scenarios were constructed from analytically synthesised phasors. After their installation, the validation chain is:

Network solver $\longrightarrow$ Waveform synthesis $\longrightarrow$ SAMPB relay pipeline $\longrightarrow$ Trip/restrain decision $+ \kappa_n$

This represents a qualitative increase in validation credibility: the relay now operates on currents that a real power system would produce, not on idealised synthetic signals.

The document is organised as follows:

- §2 — the enabling toolchain and its role
- §3 — how each Phase-7 TR evolved, with major changes
- §4 — journal paper landscape (Papers A–H)
- §5 — identified gaps (evidence, architectural, theoretical)
- §6 — detailed thesis modification plan (per chapter)
- §7 — future research objectives with timeline
- §8 — summary scorecard

2 Enabling Toolchain

Table 1: Three external simulators installed at the start of Phase 7.

Tool	Interface	Physical role	File
pandapower	Python API, <code>pn.case39()</code>	Z-bus network analysis, IEEE 39-bus case, load-flow	TF
ANDES	Python API, <code>andes.run()</code>	Positive-sequence dynamic simulation (WTDTA1, PVD1)	TF
OpenDSS	<code>opendssdirect.py</code>	Distribution-level 3-phase network with DG infeed	TF

2.1 ANDES — Dynamic IBR Simulation

ANDES provides a positive-sequence phasor-domain dynamic model of the power system. For the SAMPB validation it was used with two built-in models:

- **WTDTA1** (Type-3 DFIG wind turbine): rotor dynamics, RSC current control, crowbar activation. Used in TR-56 to validate the 10-parameter piecewise-ODE fault current model.

- **PVD1** (Solar PV converter): SRF dq current controller, zero-DC output. Used in TR-62 to validate the 2-parameter PV model.

TR-59 revealed an important limitation: ANDES reconstructs three-phase waveforms from positive-sequence states and cannot reproduce the DC offset transient with the physical fidelity required by the SAMBP LM estimator. This finding is documented as a correct engineering outcome and justifies using PSCAD/EMTDC for the final EMT validation (TR-67).

2.2 OpenDSS — Distribution Network

OpenDSS was used in TR-65 to model an 11 kV radial feeder with one distributed generator. The load-flow solution provides three-phase phasor currents at each bus; these are passed to the SAMBP 87B pipeline after DC-offset synthesis:

$$i(t) = \sqrt{2}|I| \left(\sin(\omega t + \theta) - \sin(\theta) K_{\text{DC}} e^{-t/\tau} \right), \quad \theta = \angle I + \frac{\pi}{2}.$$

2.3 pandapower — Transmission Network

pandapower was used in TR-66 to load the IEEE 39-bus New England test system (`case39()`) and compute the bus impedance matrix via:

$$\mathbf{Y}_{\text{bus}} = \text{makeYbus}(\dots), \quad \mathbf{Z}_{\text{bus}} = \mathbf{Y}_{\text{bus}}^{-1}, \quad I_f^{(k)} = \frac{V_{\text{pre}}^{(k)}}{Z_{kk}}, \quad \Delta \mathbf{V} = -\mathbf{Z}_{\bullet k} I_f^{(k)}.$$

All nine SAMBP protection functions were then evaluated against 12 fault scenarios derived from this Z-bus analysis.

3 Phase-7 Technical Report Evolution

3.1 TR-56 — DFIG Piecewise-ODE Model for 87L/87G

3.1.1 Problem

The 4-parameter SG reduced model assumes a smooth exponential DC decay. DFIG fault current has a slip-frequency oscillation during the RSC-active phase that violates this assumption, followed by an abrupt transition when the crowbar fires and bypasses the rotor circuit.

3.1.2 Contributions

- **10-parameter piecewise model:** RSC-active phase (amplitudes I_0^{RSC} , slip frequency ω_s , decay τ_{RSC} , phase ϕ_{RSC}) joined to post-crowbar phase (4 SG-equivalent params) via smoothed Heaviside with sigmoid width σ_H .
- **Analytical Jacobian:** all 10 partial derivatives in closed form, reducing LM iteration cost vs. finite-difference by $\sim 60\%$.

- **CUSUM crowbar detector:** band-power ratio test detects the RSC-to-crowbar transition in $O(1)$ sample latency.
- **Two-pass LM:** Pass 1 fits all 10 params; Pass 2 refines ω_s and τ_{DC} on the tail window where the signal-to-noise ratio is highest for frequency estimation.

3.1.3 ANDES Validation (6/6)

Validated against ANDES WTDTA1 (Type-3 DFIG) with 6-case matrix: 3 external (correctly restrained) + 3 internal (correctly tripped at ~ 950 ms). Condition number $\kappa_n \leq$ threshold confirmed in all internal cases.

3.1.4 Major evolution vs. prior SG model

The SG model is a single 4-parameter exponential. The DFIG model is a structural extension: a piecewise ODE with a real-time state-machine (CUSUM) detecting the regime transition. The CUSUM module has no equivalent in the 87T or 87B chains.

3.1.5 Status and pending work

- ✓ ANDES validation (6/6 correct)
- ❖ 10-scenario deterministic sweep (driver script not yet written)
- ❖ 1000-trial Monte Carlo (κ_n distribution table)
- ❖ Thesis §6.1 write-up

3.2 TR-57 — Bayesian SPRT for 87T IBR Inrush Discrimination

3.2.1 Problem

Second-harmonic restraint (IEC 60255-187-3 baseline) fails for IBR-fed transformers because current controllers suppress harmonics; GFM inverter inrush waveforms lack the second-harmonic signature that conventional 87T relays depend on. This causes either (a) false trips on energisation, or (b) failure to distinguish true inrush from IBR-sourced internal faults.

3.2.2 Contributions

- **Waveform asymmetry index A_k :** ratio of positive-half-cycle area to negative-half-cycle area per fundamental cycle. Inrush: $A_k \gg 1$ (asymmetric); internal fault: $A_k \approx 1$ (symmetric); CT saturation: A_k moderately elevated on alternate half-cycles.
- **Jiles–Atherton (JA) core model:** derives the theoretical distribution of A_k under inrush (H_0), giving μ_0, σ_0 calibrated to physical core parameters ($M_s, a, \alpha, c, k_{JA}$).
- **IBR current-controller model:** derives the distribution of A_k under internal fault (H_1). Current-controlled IBR suppresses DC, so $A_k \approx 1$; gives μ_1, σ_1 .
- **CT saturation model:** derives A_k under external fault with CT saturation (H_2). Flat-top on alternate half-cycles yields moderate asymmetry; gives μ_2, σ_2 .

- **Three-hypothesis SPRT:** parallel log-likelihood ratio tests H_0 vs. H_1 and H_0 vs. H_2 using Wald boundaries. Expected decision in 3–5 cycles vs. $O(\infty)$ blocking for conventional 2nd harmonic.
- **Patent claims:** four independent claims identified (asymmetry index method, three-hypothesis SPRT, CT-saturation veto, IBR parameter adaptation). Most commercially disruptive TR in Phase 7.

3.2.3 Status and pending work

- ✓ Full theoretical derivation (JA model, SPRT formulation, KL divergence)
- ❖ 24-scenario deterministic simulation
- ❖ 5000-trial Monte Carlo
- ❖ PSCAD/EMTDC validation with detailed JA transformer model
- ❖ IEEE TPWRD paper draft

3.3 TR-58 — Generalised EAC + Conservative CUEP for Relay 78

3.3.1 Problem

Relay 78 (out-of-step protection) uses the Equal Area Criterion (EAC) from classical SG swing equations. In a hybrid SG + GFM + GFL system: GFM IBR adds virtual inertia (increases decelerating area), while GFL clips current at rated (reduces decelerating area during fault clearing). The standard CUEP ignores both effects, producing a blinder setting that either trips stable power swings or fails to trip unstable ones.

3.3.2 Contributions

- **Extended power-angle equation:** augments the classical $P_e(\delta)$ with GFM virtual inertia H_{IBR} and GFL current-limited injection term.
- **Lyapunov energy function:** $V(\delta, \omega) = \sum_i (H_i/H_{\text{total}}) \frac{1}{2} \omega^2 + W(\delta)$ giving the augmented potential well for the hybrid system.
- **Conservative CUEP lower bound:** $\delta_{\text{CUEP, LB}} = \delta_{\text{CUEP, classical}} - \Delta\Omega_{IBR}$, where $\Delta\Omega_{IBR}$ is the IBR-driven reduction in potential energy.
- **IBR-corrected blinder:** $\delta_{\text{blinder}} = \delta_{\text{LB}} - 5^\circ$ (explicit conservatism margin).
- **Worked numerical example:** 30% GFL penetration; blinder crosses the impedance trajectory at $Z(\delta_{\text{LB}}) - 5^\circ$ before the classical blinder does.

3.3.3 Status and pending work

- ✓ Full theoretical derivation and Lyapunov proof
- ✓ Worked numerical example (TikZ figure produced)
- ❖ 18-scenario deterministic simulation (3 GFM $\times$ 3 GFL $\times$ 2 fault types)
- ❖ IBR parameter sensitivity sweep
- ❖ IEEE TPWRS paper draft

3.4 TR-59 — ANDES Positive-Sequence Validation of SMBP Sync-OC

3.4.1 Problem

Before TR-67 (EMT HIL), can ANDES positive-sequence simulation serve as an intermediate validation of the SMBP sync-OC estimator?

3.4.2 Findings

The 10-case matrix (5 fault types $\times$ 2 SG operating points) shows that the SMBP confidence gate correctly rejects ANDES waveforms with $\gamma_{\text{conf}} = 0.42\text{--}0.65$ (threshold 0.70). This is a *correct* engineering outcome: the positive-sequence reconstruction lacks the DC offset physics that the LM estimator needs to identify τ_{DC} and I_{DC} .

For comparison, synthetic data gives $\gamma_{\text{conf}} = 0.89\text{--}0.97$.

3.4.3 Major finding

ANDES positive-sequence simulation is *not* suitable for protection-grade validation of the SMBP estimator. This finding:

1. Justifies TR-67 (RTDS/PSCAD) as the definitive validation path.
2. Proves the SMBP confidence gate is functioning correctly as a quality filter.
3. Establishes that the validation chain must use a true EMT simulator for final results.

3.4.4 Status

- ✓ 10-case validation matrix
- ✓ Confidence score breakdown and physical interpretation
- ❖ PDF compilation (LaTeX source exists, compile pass pending)
- ❖ EMT follow-up (TR-67)

3.5 TR-62 — Solar PV 2-Parameter Model and k_{ibr} -Driven Selector for 87L

3.5.1 Problem

The *ieee_line_diff* paper validates 87L only for SG sources. PV inverters inject near-pure sinusoids: the SRF current controller tracks d -axis current to rated with negligible DC. When the 4-parameter SG model is applied to a PV fault, the two DC terms are collinear (both approach zero) and the Jacobian becomes rank-deficient ($\kappa_n \rightarrow 10^6$).

3.5.2 Contributions

- **2-parameter PV model:** $\theta_{\text{PV}} = [I_{\text{fund}}, \phi_{\text{fund}}]$; DC terms fixed to zero analytically. Jacobian rank = 2, $\kappa_n \leq 1.00$.

- **DC Decay Ratio (DDR):** $DDR = (i_{\text{peak}} - i_{\text{ss}})/i_{\text{peak}}$ measured in the first 20 ms post-fault. $DDR < 0.05$: PV model. $DDR \in [0.05, 0.20]$: DFIG model. $DDR > 0.20$: SG model.
- **Exponential veto:** if fitted $\tau_{\text{DC}} > 200$ ms $\Rightarrow$ override to PV model (prevents over-parameterisation on near-sinusoidal faults).
- **Drop-in replacement architecture:** unified LM call with DDR selector switching the parameter vector before each LM step; the 87L pipeline is unchanged.

3.5.3 ANDES Validation (6/6)

Validated against ANDES PVD1 with 6-case matrix: 3 external (correctly restrained) + 3 internal (tripped). $\kappa_n = 1.00$ in all internal cases — perfect identifiability. DDR correctly selects PV model in 100% of cases.

3.5.4 Status and pending work

- ✓ 2-parameter model derivation and Jacobian
- ✓ ANDES PVD1 validation (6/6)
- ❖ 9-scenario deterministic PV study (driver script)
- ❖ 5000-trial Monte Carlo (P_D , t_{50} , t_{95} , DDR accuracy)
- ❖ *ieee_line_diff* paper §III update (add PV results column)

3.6 TR-65 — OpenDSS 87B Bus Differential, 11 kV Distribution Bus

3.6.1 Problem

All prior 87B validation (TR-04) used synthetic phasors. This does not demonstrate that the SAMBP 87B pipeline operates correctly when the bus currents are computed by a real network solver under distribution-level topology with DG infeed, line charging, and CT mismatch.

3.6.2 What was built

- OpenDSS 11 kV radial feeder: VSOURCE 11 kV, 3 feeders, 1 DG at Feeder 3.
- Physics-correct DC offset from OpenDSS phasor output: $\theta = \angle I + \pi/2$, sign from $-\sin \theta$.
- CT model: $\epsilon_{\text{CT}} = I_{\text{sat}}/(1 + I_{\text{sat}}/I_{\text{peak}})$.
- 6 scenarios: normal load, external F1, external F2 + CT-sat, internal AG, internal 3-phase, external F2 + DG.

3.6.3 Results (6/6)

All six scenarios returned the correct relay decision. The external-fault + CT-saturation case was correctly vetoed by the Stage-2 gate ($\kappa_n = 4.9$, $f_{\text{int}} = 0.45 < 0.60$). The DG reverse-infeed case correctly balanced KCL at the bus.

3.6.4 Status and pending work

- ✓ 6/6 correct decisions
- ✓ Stage-2 veto on CT saturation confirmed
- ❖ Multi-feeder buses (3-winding, bus-section switch)
- ❖ Monte Carlo study (CT mismatch, DG penetration sweep)
- ❖ HIL integration (TR-67)

3.7 TR-66 — IEEE 39-Bus Full-System Validation (All 9 Functions)

3.7.1 Problem

No single previous test had exercised all nine SAMBP protection functions simultaneously on a realistic multi-machine transmission system. Each function had been validated in isolation; their ability to *co-exist without cross-tripping* was unproven.

3.7.2 Network

IEEE 39-bus New England test system (pandapower case39()): 39 buses, 35 lines, 11 transformers, 9 generators + external grid, 100 MVA / 345 kV base. Z-bus fault analysis provides $I_f^{(k)} = V_{\text{pre}}^{(k)} / Z_{kk}$ at each faulted bus.

3.7.3 Nine protection functions

Element	Function	Key parameter
51G	Generator overcurrent (inverse-time)	$I_{\text{pickup}} = 1.1 \text{ p.u.}$, TMS = 0.5
87G	Generator differential (two-terminal zone)	Slope 20%, pickup 0.2 p.u.
87B	Bus differential (SAMBP pipeline)	Slope 30%, $\kappa_n < 30$, f_{int} gate
87L	Line differential (SAMBP pipeline)	$k_{\text{line}} = 0.12 \text{ p.u. threshold}$
87LN	Negative-sequence differential	$I_2 > 0.06 \text{ p.u. threshold}$
87LN0	Zero-sequence differential	$I_0 > 0.04 \text{ p.u. threshold}$
TDCS	IBR adaptive threshold	$K_\sigma = 3$, $\delta_{\text{floor}} = 0.025 \text{ p.u.}$
87T	Transformer differential (SAMBP pipeline)	Yd11, slope 15/25%, $\kappa_n < 30$ gate
21	Mho distance (3-zone)	Zone1 = 80%, Zone2 = 120%, Zone3 = 180%

3.7.4 Key engineering decisions resolved

87G convention. For an external bus fault, the HV terminal current flows *out* of the generator zone (convention: into-zone is positive). Therefore:

$$I_{\text{op}}^{87G} = |I_{\text{stator}} + (-I_{\text{delivered}})| = 0 \quad (\text{external fault}) \quad I_{\text{op}}^{87G} = |I_{\text{stator}}| \gg 0 \quad (\text{internal fault}).$$

Using $+I_{\text{delivered}}$ (wrong sign) produced false trips on 4 of 12 scenarios.

KCL balance at Bus 39. Generator G9 contributes only $0.39/15.03 = 2.6\%$ of the Bus 39 fault current. The remaining 97.4% is supplied via Line 9-39 from the network. Ignoring the correct proportional split created a 5.72 p.u. false differential.

Yd11 cancellation. For a transformer external fault, the LV phasor must lead the HV phasor by $5\pi/6$ so that after the M_X compensation matrix, the restrained differential is zero. Using the same phase (LV = HV) produced a 41 p.u. false trip.

TDCS IBR detection. The IBR scenario produces $k_{ibr} = 0.08$ p.u. < 0.12 p.u. (87L threshold), so 87L correctly restrains. TDCS post-RMS = 0.057 p.u. $>$ threshold 0.026 p.u.; TDCS alone trips, demonstrating the IBR fallback function.

3.7.5 Results

Scenario	Expected trip(s)	Outcome
Normal load	None	Done
Bus 39 — 3-phase	87B	Done
Bus 39 — A-G	87B	Done
Gen stator — 3-phase	51G, 87G	Done
Line 9–39 — 3-phase	87L, TDCS	Done
Line 9–39 — A-G	87L, 87LN, 87LN0, TDCS	Done
Line 9–39 — IBR	TDCS only	Done
Bus 9 external — 3ph	None (Zone 2 delayed)	Done
Line 8–9 — Zone 1	21	Done
Line 8–9 external	None	Done
Trafo internal — 3ph	87T	Done
Trafo external — 3ph	None	Done
Overall selectivity		12/12 = 100%

4 Journal Paper Landscape

Table 2: Eight journal papers in the SMBP programme.

Label	Title	Target journal	Status	Chapter
Paper A	Real-Time Inverse Parameter Estimation for Adaptive Overcurrent	IEEE TPWRD	Draft	Ch. 3
Paper B	Hilbert–Fortescue Sequence Estimation and Cascaded OC Coordination	IEEE TPWRD	Draft	Ch. 3
Paper C	Self-Adaptive Model-Based Protection: A Unified Framework	IEEE TPS	Draft	Ch. 7

Label	Title	Target journal	Status	Chapter
Paper D	Validation Boundaries of Reduced-Order Inverse Protection Under 6th-Order Park dq Dynamics	IEEE TEC	Draft	Ch. 2
Paper E	Impedance-Aware Adaptive Very-Inverse OC for High-Impedance Faults	IEEE TPWRD	Draft	Ch. 3
Paper F	Sequence-Component Inverse Estimation for Adaptive OC Under Asymmetric Faults	IEEE TPWRD	Draft	Ch. 7
Paper G	Coordinated Adaptive OC for Parallel Synchronous Generators Using Online Inverse Estimation	IEEE TPWRD	Draft	Ch. 3
Paper H	IBR-Adaptive Line Differential Protection: Harmonic Discrimination, Zero-Seq, TDCS	IEEE TIE	Draft	Ch. 4

Papers A–H are all in draft form. The most urgent updates are to Paper C (needs TR-66 selectivity matrix) and Paper H (needs TR-62 PV results).

5 Identified Gaps

5.1 G1 — Simulation Evidence Gaps in Existing TRs

Table 3: Missing simulation runs blocking paper submissions.

TR	Missing evidence	Impact
TR-56	10-scenario deterministic sweep; 1000-trial MC; κ_n table empty	Ch. 6 §6.1 and IEEE TEC paper blocked
TR-57	24-scenario sim and 5000-trial MC not run; no PSCAD validation	IEEE TPWRD paper blocked
TR-58	18-scenario simulation not run; blinder performance unquantified	IEEE TPWRS paper blocked
TR-59	No PSCAD/EMT follow-up; positive-sequence limitation documented but not resolved	TR-67 is the only path forward
TR-62	9-scenario PV study not run; DDR accuracy not measured empirically	Paper H §III update blocked
TR-65	MC, multi-feeder, 3-winding bus not done	Single-feeder result only
TR-66	Multi-corridor extension (only Bus 39 covered); 10k-trial MC pending	Paper C §IV update and Chapter 7 blocked

5.2 G2 — Missing TRs from the Gap Plan

Table 4: Five TRs planned but not yet written.

TR	Topic	Gap filled
TR-60	Relay 40 (Loss of Excitation) with GFM admittance correction	No LOE treatment anywhere
TR-61	64G stator earth fault, optimal sub-harmonic injection $f^* = 25\text{--}35$ Hz	No stator EF treatment
TR-63	Relay 81 IFDI for zero-inertia GFM networks	Frequency detection entirel
TR-64	IBR-aware 87T with SPRT + 5k MC	TR-57 is theory; TR-64 is mentation
TR-67	HIL with DFIG and PV emulators on RTDS/PSCAD	Whole SMBP stack valid

5.3 G3 — Paper Update Gaps

Table 5: Papers requiring material updates from new TR results.

Paper	What needs updating	Reason
Paper H	PV 2-param model (§III), DDR selector (§IV), PV results column (§V)	TR-62 adds third source type
Paper C	Ch. 7 integration results from TR-66 selectivity matrix	TR-66 provides cross-element evidence
Paper D	Validity boundary for DFIG and PV sources absent	TR-56/TR-62 change the boundary conditions
Papers A/E/F/G	IBR infeed OC coordination section absent	No paper addresses IBR effect on 51 settings

5.4 G4 — Architectural Gaps

1. **No hardware validation.** All validation is Python simulation. SMBP claims real-time capability but has never run on embedded hardware or an RTDS.
2. **Single protection corridor.** TR-66 validates one corridor (Bus 39). The 39-bus system has 10 generators and 35 lines; the remaining 9 corridors are untested.
3. **No adaptive setting update loop.** SMBP estimates parameters and decides trip/restrain but never feeds κ_n or model type back to setting management. The “self-adaptive” claim in Paper C is not demonstrated end-to-end.
4. **No communications model.** 87L assumes perfect latency. The interaction between TDCS and channel degradation (packet loss, jitter) has not been co-validated.
5. **Sequence elements not in unified test.** 87LN and 87LN0 trip correctly in TR-66 (line9_39_ag scenario), but TR-22/TR-25 (the defining TRs) have not been updated to show IBR-sourced AG fault behaviour.

5.5 G5 — Theoretical Gaps

1. **DFIG sub-synchronous resonance (SSR).** TR-56 models crowbar transition but not SSR coupling between the DFIG shaft and series-compensated lines — a known failure mode for 87L protection.
2. **GFM virtual inertia during fault ride-through.** TR-58 assumes GFM tracks SG angle during swing. Grid-forming converters with current limiters enter a different regime during fault; the EAC energy function is not valid during current limiting.
3. **Multi-infeed IBR coordination.** All TRs model one IBR source. A bus simultaneously fed by DFIG and PV has a mixed DDR; the TR-62 selector has no defined behaviour for this case.
4. **Non-stationary noise in TDCS.** TDCS uses a static σ_{pre} . If background load switches near the IBR bus (motor starts, capacitor energisation), σ_{pre} evolves. No adaptive noise floor update is defined.

6 Thesis Modification Plan

6.1 Chapter 2 — Machine Modelling

Current state: 6th-order Park dq truth model (Paper D), reduced 4-parameter SG model.

- **Add §2.4 — DFIG Reduced Piecewise-ODE Model** (from TR-56 §3–§5). Present the 10-parameter model as a natural structural extension of the 4-parameter SG model, framed as: “as the source transitions from SG to DFIG, the single exponential is replaced by a two-regime ODE.”
- **Add §2.5 — PV 2-Parameter Model** (from TR-62 §2). Show the model reduction as a motivated collapse: when $\tau_{\text{DC}} \rightarrow \infty$, the DC terms vanish and the identifiable parameter space collapses to $[I_{\text{fund}}, \phi_{\text{fund}}]$.
- **Update validity boundary discussion.** $L^* \approx 150$ ms for SG (AVR time constant). For DFIG: boundary set by crowbar transition time (~ 50 – 100 ms). For PV: no DC $\Rightarrow$ no validity boundary. A three-row comparison table (SG vs. DFIG vs. PV) is required.
- **Add Figure 2.X:** source-model taxonomy tree (SG $\rightarrow$ 4-param; DFIG $\rightarrow$ 10-param piecewise; PV $\rightarrow$ 2-param zero-DC).

6.2 Chapter 3 — Overcurrent Protection (51)

Current state: Papers A, B, E, F, G covering SG feeders, HIF, sequence decomposition, cascade coordination, and parallel generator coordination.

- **Add §3.6 — IBR infeed effect on 51 coordination.** When an IBR is on the same bus as the protected SG, the fault current at the relay is $I_{\text{relay}} = I_{\text{SG}} + k_{\text{ibr}} I_{\text{rated}}^{\text{IBR}}$. The existing Very-Inverse characteristic uses only SG-sourced current in its $M = I/I_{\text{pickup}}$ ratio. An IBR infeed correction factor is needed.

- **Update Papers A and G selectivity tables.** Re-run the 10-scenario and multi-gen scenarios adding a PV or DFIG at the feeder bus. Show that existing coordination holds when $k_{ibr} \leq 0.30$ p.u..
- **Update Paper E (HIF).** Add a note that HIF detectability with IBR infeed may be reduced: the rising impedance signature can be masked when the IBR current controller clamps output.

6.3 Chapter 4 — Line Differential (87L)

Current state: Paper H covers 21 scenarios for SG source only.

- **Add §4.5 — PV-Source 87L via 2-Parameter Model and DDR Selector.** The DDR selector algorithm (TR-62 §5) must be promoted from TR appendix to thesis body — it is the first automated source-type classifier in the SAMPB chain and deserves a dedicated algorithm box.
- **Update Paper H results table.** Add 9 PV scenarios from TR-62 Table 5. Column header “Source type: SG” becomes a three-row grouping: SG, DFIG, PV.
- **Add §4.6 — TDCS with non-stationary background.** Frame TR-24 (adaptive TDCS) as the reference and identify the static σ_{pre} as a known limitation.
- **Update §4.3 (sequence elements).** Cross-reference TR-66 §5.5: 87LN trips at $I_2 = 4.88$ p.u. (AG fault on line9_39), confirming the 0.06 p.u. threshold is conservative by $80\times$.

6.4 Chapter 5 — Transformer Differential (87T)

Current state: TR-03 (87T foundation), TR-57 (SPRT theory only). No validated simulation results exist.

- **Write §5.1–§5.3 from TR-57 theory.** JA core model, asymmetry index, and SPRT formulation are publication-quality derivations and go into the chapter body directly.
- **§5.4 must explicitly state results are PENDING (TR-57a).** Do not write a results section without data.
- **§5.5 — Integration with SAMPB 87T gate** (TR-57 §6). The SPRT replaces 2nd-harmonic blocking but is *additive* to the Stage-2 κ_n gate — both must pass for a trip.
- **Critical path:** This chapter is on the Q3 2026 target. TR-57 simulation must be the next coding task after TR-62a.

6.5 Chapter 6 — Generator Protection Suite (40/64G/78/81/87G)

Current state: TR-56 (DFIG 87G/87L model), TR-58 (Relay 78 EAC). TRs for 40/64G/81 not yet written.

- **§6.1 — DFIG machine model (TR-56):** write from TR-56 §2–§8, ending with the 6/6 ANDES validation.
- **§6.2 — Generator differential 87G:** note that TR-66 §3.2 establishes the two-terminal zone convention. Write this as the governing convention for all 87G analysis in the chapter.
- **§6.3 — Relay 78 (out-of-step):** write from TR-58 §3–§5. The Lyapunov energy function and conservative CUEP bound are the theoretical centrepiece of this section.
- **§6.4–§6.6 PENDING:** Relay 40, 64G, 81 require TR-60, TR-61, TR-63. Mark with explicit PENDING placeholders and target dates.
- **Submission target:** Chapter 6 cannot be submitted until Q1 2028.

6.6 Chapter 7 — System Integration

Current state: Paper C (unified SAMBP framework), Paper F (sequence estimation). TR-65 and TR-66 are new evidence not yet reflected in these papers.

- **Add §7.4 — IEEE 39-Bus Multi-Function Coordination (TR-66).** The selectivity matrix (TR-66 Table 6) is the primary evidence that all 9 functions coexist without false interaction at transmission level.
- **Add §7.5 — OpenDSS Distribution-Level Integration (TR-65).** Connects transmission protection (TR-66) to distribution protection (TR-65) — a unique two-level validation across voltage levels.
- **Update Paper C §IV.** The unified framework claim is now supported by 12/12 selectivity. The current Paper C has no multi-function, multi-source evidence; TR-66 fills this gap directly.
- **§7.6 — Remaining coordination gaps.** Multi-corridor extension (35 lines not covered) and HIL (TR-67). State as future work with specific quantitative targets.

6.7 Chapter 8 (new) — Conclusions and Future Directions

- Section on RTDS/HIL (TR-67 scope and timeline)
- Section on adaptive setting update loop (the missing link in the “self-adaptive” claim of Paper C)
- Section on multi-infeed IBR coordination
- Explicit statement on out-of-scope items: LVRT/HVRT control, PV string internal faults, cybersecurity, machine learning, economic optimisation.

7 Future Research Objectives

7.1 Near-Term (Q2–Q3 2026): Close Evidence Gaps

1. **TR-57a** — Run 24-scenario 87T SPRT deterministic study. Compute KL divergence empirically and compare against theoretical ESC formula. *This is the highest-priority task: it unblocks the most commercially disruptive paper and the Chapter 5 critical path.*
2. **TR-56a** — Write `sampb_generator` driver. Run 10-scenario sweep + 1000-trial MC. Populate κ_n distribution table.
3. **TR-62a** — Run 9-scenario PV deterministic study and 5000-trial MC. Measure DDR accuracy vs. source type (P_D , t_{50} , t_{95}).
4. **Compile TR-59 PDF** — LaTeX source exists; run `pdflatex + bibtex`.

7.2 Medium-Term (Q4 2026–Q1 2027): Generator Protection Suite

5. **TR-60** — Relay 40 LOE with GFM admittance correction. Key equation: $Z_{40} = E_q^2 / P - j(X_d - jX_q/2)$ corrected for GFM virtual admittance. Requires a new distance-type element not in current SAMPB code.
6. **TR-61** — 64G stator earth fault with optimal sub-harmonic injection. Optimal frequency $f^* = 25\text{--}35$ Hz minimises capacitive coupling noise while maximising earth-fault sensitivity.
7. **TR-64** — Complete IBR-aware 87T with SPRT + 5000-trial MC. Extends TR-57 from theory to validated implementation.

7.3 Medium-Term (Q2–Q3 2027): System-Level Completion

8. **TR-63** — Relay 81 IFDI for zero-inertia GFM networks. Requires a frequency estimation algorithm robust to rapidly changing virtual inertia — a genuinely novel problem with no published solution.
9. **Multi-corridor TR-66 extension** — Extend IEEE 39-bus validation to all 10 generator corridors. Generalise Z-bus superposition code for arbitrary bus fault locations.
10. **Paper H update** — Merge TR-62 PV material into *ieee_line_diff* §III–§V. Target IEEE TIE submission Q3 2027.
11. **Paper C update** — Merge TR-66 selectivity matrix into §IV. Reframe contribution as “multi-function, multi-source, multi-level integration.”

7.4 Long-Term (Q4 2027–Q2 2028): Hardware Validation and Self-Adaptation

12. **TR-67** — RTDS/PSCAD HIL with real DFIG and PV emulators. Definitive validation closing the gap TR-59 identified: positive-sequence ANDES is insufficient for protection-grade validation.

13. **Adaptive setting update loop** — Implement feedback from the LM estimator’s κ_n output to the 51 relay’s I_{pickup} . When κ_n rises, widen the 51 time-grading margin automatically. This makes “self-adaptive” a literal truth, not a framing claim.
14. **Multi-infeed DDR mixer** — Define DDR behaviour for a bus fed by both DFIG (DDR $\in [0.05, 0.20]$) and PV (DDR < 0.05) simultaneously. Requires a convex combination rule or parallel hypothesis test.
15. **Non-stationary TDCS** — Replace static σ_{pre} with an EWMA adaptive estimate. Makes the threshold adaptive to motor starts, capacitor bank energisation, and IBR MPPT transients.

8 Summary Scorecard

Table 6: Before/after comparison across key research dimensions.

Dimension	Before Phase 7	After Phase 7
Source types modelled	SG only	SG, DFIG, PV (3 types)
Protection elements tested in isolation	All 9 elements	All 9 simultaneously (TR-66)
Network realism	Synthetic phasors	OpenDSS (distribution) + pandapower/Z-bus (transmission)
Dynamic simulator	None	ANDES (WTDTA1 + PVD1)
Validation scenarios (cumulative)	~55	~91
Phase-7 PDFs compiled	—	6 (TR-56/57/58/62/65/66)
Papers with IBR content	1 (Paper H draft)	1 complete + 2 pending updates
Pending simulation runs	0	5 (TR-56a, 57a, 58, 62a, 66 ext.)
Overall selectivity (best test)	7/7 (87T, TR-04)	12/12 (all 9 functions, TR-66)

Critical path: TR-57a simulation (Q2 2026) → TR-56a MC (Q3 2026) → TR-62a MC (Q3 2026) → TR-60/61/64 (Q1 2027) → TR-63/67 (Q4 2027) → Chapter 7 write-up → Submission **Q3 2028**.

The most urgent single action is **TR-57a simulation** — it unblocks the most commercially significant patent and the only paper targeting IEEE TPWRD on transformer differential protection. The second most urgent is **TR-62a** — the data is needed to update Paper H before the IEEE TIE submission.

A Technical Report Directory

Table 7: All Phase-7 technical reports with file locations and PDF status.

TR	Title (short)	LaTeX file	PDF
TR-56	DFIG piecewise-ODE model	TR56_DFIG_.../main_report56.tex	✓
TR-57	SPRT 3-hypothesis 87T	TR57_SPRT_.../main_report57.tex	✓
TR-58	Generalised EAC + Relay 78	TR58_EAC_.../main_report58.tex	✓
TR-59	ANDES sync-OC validation	TR59_ANDES_.../main_report59.tex	✗
TR-62	PV 2-param + DDR selector	TR62_PV_.../main_report62.tex	✓
TR-65	OpenDSS 87B distribution	TR65_OpenDSS_.../main_report65.tex	✓
TR-66	IEEE 39-bus, 9-function	TR66_IEEE39_.../main_report66.tex	✓

B Code and Output Locations

ANDES DFIG validation (TR-56): 04_code/sampb/generator/

ANDES PV validation (TR-62): 04_code/sampb/line_diff/ (PV extension modules)

OpenDSS 87B script (TR-65): 04_code/sampb/system/run_tr65_opendss.py

IEEE 39-bus script (TR-66): 04_code/sampb/system/run_tr66_ieee39bus.py

TR-66 results CSV: 04_code/sampb/system/outputs/tr66/tr66_results.csv

All TR LaTeX sources: 03_technical_reports/phase_7_IBR_extension/

All journal paper sources: 02_papers/